

4.4 Indeterminate Forms and L'Hôpital's Rule

In this section we will discuss a technique for evaluating limits of indeterminate form.

What is an indeterminate form?

Can you think of other similar examples (i.e. involving 0 and/or ∞) that are NOT indeterminate?

Examples of Indeterminate Forms

- $\lim_{x \rightarrow 0} \frac{e^x - 1}{\sin(x)}$
- $\lim_{x \rightarrow 1} \frac{\ln x}{x - 1}$
- $\lim_{x \rightarrow \infty} \frac{x^2 + 1}{x^3 + 2x}$
- $\lim_{x \rightarrow \infty} \frac{x}{e^x}$
- $\lim_{x \rightarrow \infty} (\sqrt{x^2 + 7} - x)$
- $\lim_{x \rightarrow \infty} \left(1 + \frac{4}{x}\right)^x$
- $\lim_{x \rightarrow 0^+} x^{\frac{1}{\ln x}}$

What are some methods you learned to evaluate limits in your previous calculus course?

Example 1

Which of the following limits results in an indeterminate form? Select all that apply.

1. $\lim_{x \rightarrow \infty} \frac{x^2 + 1}{e^{-x}}$

3. $\lim_{x \rightarrow 0^+} \ln(x) + \frac{1}{x}$

2. $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$

4. $\lim_{x \rightarrow \infty} x^x$

L'Hôpital's Rule

Suppose that $f(a) = g(a) = \underline{\hspace{1cm}}$, that f and g are
 $\underline{\hspace{2cm}}$ on an open interval I containing a , and that
 $\underline{\hspace{2cm}}$ on I if $x \neq a$. If

$$\begin{array}{lcl} \lim_{x \rightarrow a} f(x) = 0 & & \lim_{x \rightarrow a} f(x) = \pm\infty \\ \lim_{x \rightarrow a} g(x) = 0 & \text{or} & \lim_{x \rightarrow a} g(x) = \pm\infty \end{array}$$

then

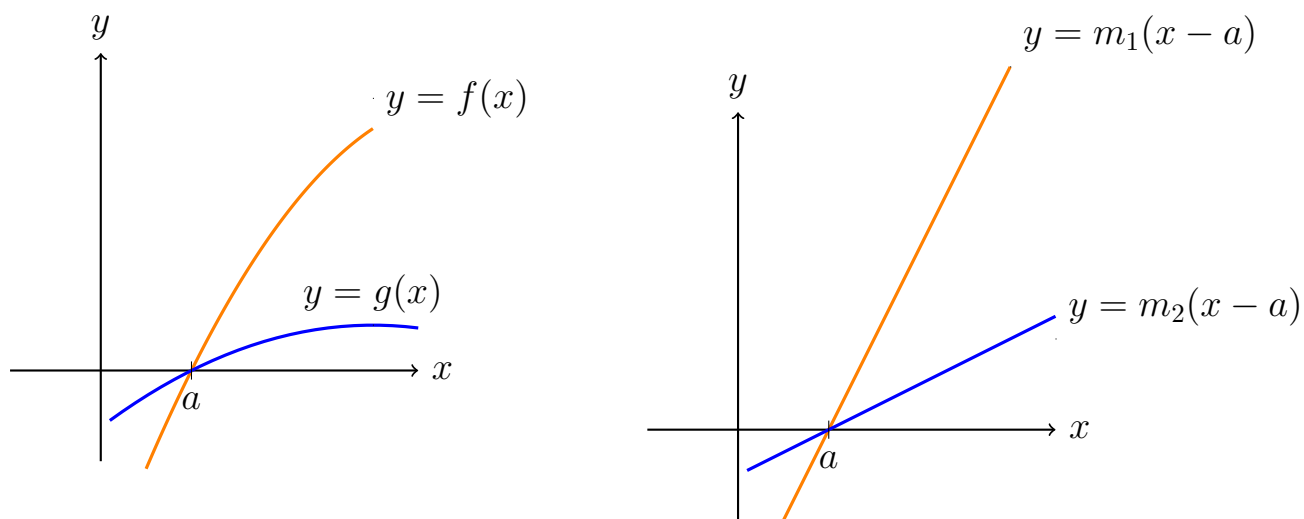
$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} =$$

assuming that the limit on the right side of this equation exists.

Notes

- We can apply L'Hôpital's Rule to one-sided limits.
- Before attempting to apply L'Hôpital's Rule
 - check that the limits of $f(x)$ and $g(x)$ exist before attempting to apply L'Hôpital's Rule.
 - determine which indeterminate form you have
- If $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ does not exist, then $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ still *might* exist.

Why does L'Hôpital's Rule work? Let's examine it geometrically!



Sometimes we approximate a curve using its tangent line.

What do you notice about the relationship between $y = f(x)$ and $y = g(x)$ and $y = f'(x)$ and $y = g'(x)$?

Now let's prove it!

Example 2

Evaluate the following limits using L'Hôpital's Rule when appropriate:

1. $\lim_{x \rightarrow 0} \frac{e^x - 1}{\sin(x)}.$

2. $\lim_{x \rightarrow 1} \frac{\ln x}{x - 1}$

3. $\lim_{x \rightarrow \pi} \frac{\tan(x)}{\cos(x)}$

4. $\lim_{x \rightarrow \infty} \frac{x^2 + 1}{x^3 + 2x}$

5. $\lim_{x \rightarrow \infty} \frac{x}{e^x}$

6. $\lim_{x \rightarrow 0} \frac{x + \sin(x)}{x}$